IEEE STD 830 — SOFTWARE REQUIREMENTS SPECIFICATION

Deepfake Speech Detection & Biometric Authentication System

with Multimodal XAI Pipeline — NLP · Audio · Computer Vision

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Revision History

Version	Date	Author	Change Summary
v1.0	2026-03-17	Ferel, Safa	Initial SRS draft — KCVanguard brainstorm translation
v2.0	2026-03-17	Ferel, Safa	IEEE 830 restructure; Architectural Synthesis added; multimodal FR breakdown (NLP/Audio/CV); updated NFR table
v2.1	2026-03-17	Ferel, Safa	NLP module: SEA-LION replaced by Qwen 2.5; output language changed to English. Audio clip duration updated to 2 s (32,000 samples). FR-CV-014: Softmax normalisation mandated for band attribution. FR-DEP-010: ONNX Runtime escalated to SHALL/mandatory. Open Questions Q1 and Q2 closed.

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Architectural Synthesis
Pre-specification reasoning — how the three modalities technically interact

0. Architectural Synthesis

This section constitutes a step-by-step architectural reasoning exercise conducted prior to formalising requirements, as mandated by the updated prompt. It analyses how the three AI modalities — Audio/Speech (audio signal processing), Computer Vision (CNN-based spectrogram classification), and NLP (large language model explanation) — are technically coupled within the DSDBA pipeline.

0.1 How the three modalities interact

The system is best understood as a **sequential multimodal fusion pipeline** in which each modality acts as a producer for the next. It is not a parallel ensemble — the modalities are strictly ordered and each depends on the output of its predecessor.

Stage 1 — Audio DSP: The raw waveform (time-domain, 1D) is transformed via Short-Time Fourier Transform (STFT) into a 2D time-frequency Mel Spectrogram. This is a *signal processing operation*, not a learned one. It converts an audio question ("is this voice real?") into a visual question ("does this image look like real speech patterns?"), enabling a Computer Vision model to be applied.

Stage 2 — Computer Vision: EfficientNet-B4, an ImageNet-pretrained CNN, ingests the 3×224×224 spectrogram tensor and learns to discriminate bonafide from spoof spectrograms. Its convolutional feature maps encode spatial patterns in the time-frequency plane — formant continuity, periodicity artefacts, energy distribution — that are invisible in the raw waveform. The final convolutional block activations are the *interface point between the CV and XAI modules*: Gradient-weighted Class Activation Mapping (Grad-CAM) back-propagates the predicted class gradient through these activations to produce a saliency heatmap localised to the spectrogram's frequency bands.

Stage 3 — NLP (Qwen 2.5): The structured outputs of Stage 2 (label, confidence score, per-band Grad-CAM attribution percentages) are serialised into a natural-language prompt template and submitted to the Qwen 2.5 large language model. Qwen 2.5's task is purely *generation*: it receives structured data and produces a coherent, acoustically-informed explanation paragraph in English. There is no feedback from the NLP layer back to the CV or Audio layers — the pipeline is strictly feed-forward.

0.2 Critical coupling points and design risks

The following coupling points require explicit engineering decisions before implementation begins:

- **Spectrogram → tensor shape contract:** The Audio DSP module must produce exactly [3, 224, 224] float32 tensors. Any deviation (wrong n_mels, wrong duration) silently corrupts model inputs without raising an exception.
- **Grad-CAM target layer identity:** The XAI module must reference the correct named layer in torchvision's EfficientNet-B4 graph. This is not configurable at runtime without re-validation — it must be locked during training.
- **Frequency band → spectrogram row mapping:** Band attributions (0–500 Hz, 500 Hz–2 kHz, 2–4 kHz, 4–8 kHz) must be derived from the Grad-CAM heatmap using the Mel filter bank's frequency-to-bin mapping, not a naive linear slice of the 128-row tensor.
- **Qwen 2.5 API latency:** The NLP stage is the only externally-dependent, non-deterministic component. Its failure must not block the CV result from being displayed. The UI architecture must decouple Stage 3 from the display of Stage 2 results.

Section 1
Introduction

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines all functional and non-functional requirements for the Deepfake Speech Detection and Biometric Authentication System (DSDBA), version 2.1. It supersedes DSDBA-SRS-2026-001 (v1.0) and incorporates: (a) an Architectural Synthesis section per the updated specification template; (b) restructuring to align with IEEE Std 830-1998; (c) explicit breakdown of functional requirements by modality — Audio/Speech, Computer Vision, and NLP.

This document is the primary contract between the development team and all project stakeholders. All design, implementation, and acceptance testing activities SHALL be traceable to requirements defined herein.

1.2 Scope

DSDBA is a four-stage, feed-forward multimodal inference system that determines the authenticity of a speech audio sample and provides an explainable, human-readable justification in English. The four stages map directly to the three technical modalities plus a deployment layer:

Stage 1 — Audio	Waveform ingestion, resampling, segmentation, Mel Spectrogram extraction
Stage 2 — CV	EfficientNet-B4 binary classification (bonafide / spoof) + Grad-CAM XAI
Stage 3 — NLP	Qwen 2.5 LLM prompt construction and English explanation generation
Stage 4 — Deploy	Gradio/Streamlit web application on HuggingFace Spaces

Out of scope: speaker identification, real-time streaming, mobile-native applications, MLOps pipeline management, and multi-language UI.

1.3 Definitions, Acronyms, and Abbreviations

DSDBA	Deepfake Speech Detection and Biometric Authentication System
Bonafide	Authentic human speech recorded from a real speaker without synthesis or conversion
Spoof	Synthetically generated or voice-converted speech intended to impersonate a real speaker
EER	Equal Error Rate — the point at which FAR = FRR; lower is better
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
Mel Spectrogram	2D time-frequency representation of audio using the perceptual Mel frequency scale

STFT	Short-Time Fourier Transform — decomposes audio into frequency components over time
MBConv	Mobile Inverted Bottleneck Convolution — the core repeating block in EfficientNet
SE	Squeeze-and-Excitation — channel-attention mechanism embedded in each MBConv block
Grad-CAM	Gradient-weighted Class Activation Mapping — CNN explanation via gradient backprop
Qwen 2.5	Qwen 2.5 large language model by Alibaba Cloud — used for English explanation generation
FoR Dataset	Fake-or-Real audio dataset by M. Abdel-Dayem (Kaggle), used for training and evaluation
XAI	Explainable Artificial Intelligence
HF Spaces	HuggingFace Spaces — free-tier cloud deployment platform for ML applications
DXA	Device-independent XML Absolute units (1440 DXA = 1 inch) — docx internal unit
IEEE 830	IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications

1.4 References

1. Abdel-Dayem, M. (2023). The Fake-or-Real (FoR) Dataset. Kaggle.
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3. Selvaraju, R.R. et al. (2017). Grad-CAM: Visual Explanations from Deep Networks. ICCV 2017. arXiv:1610.02391.
4. Park, D.S. et al. (2019). SpecAugment: A Simple Data Augmentation Method. Interspeech 2019.
5. Alibaba Cloud. (2024). Qwen 2.5: Series of Large Language Models. huggingface.co/Qwen.
6. IEEE Std 830-1998. IEEE Recommended Practice for Software Requirements Specifications.
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1.5 Document Conventions

- **SHALL** — mandatory requirement; non-compliance = defect.
- **SHOULD** — strongly recommended; non-compliance requires justification.
- **MAY** — optional; included for completeness.
- **[Assumption]** — engineering inference where notes were silent.
- Requirement IDs: FR-AUD-### (Audio), FR-CV-### (Computer Vision), FR-NLP-### (NLP), FR-DEP-### (Deployment), NFR-###.

Section 2
Overall Description

2. Overall Description

2.1 Product Perspective

DSDBA is a greenfield standalone research prototype. It does not integrate with any existing enterprise system. It interfaces with four external entities:

- FoR Dataset (Kaggle) — primary training and evaluation corpus
- HuggingFace Model Hub — model artefact versioning and retrieval
- HuggingFace Spaces (Gradio runtime) — public deployment host
- Qwen 2.5 API (Alibaba Cloud) — NLP explanation generation

The internal pipeline is strictly sequential: Audio DSP → CV Inference → XAI → NLP Explanation. No component produces feedback to an upstream stage.

2.2 Product Functions — Summary

F-1 Audio ingestion	Accept .wav/.flac upload; resample to 16 kHz mono; segment to fixed 2-second clips
F-2 Spectrogram extraction	STFT → 128-band Mel Spectrogram → dB normalisation → [3×224×224] tensor
F-3 Deepfake classification	EfficientNet-B4 binary inference → bonafide probability score
F-4 Saliency mapping	Grad-CAM heatmap over final conv block; overlay on spectrogram PNG
F-5 Band attribution	Decompose saliency into four acoustic frequency bands → percentage scores
F-6 NL explanation	Qwen 2.5 prompt → English 3–5 sentence justification
F-7 Web interface	Upload widget, waveform/spectrogram display, verdict card, explanation panel

2.3 User Classes and Characteristics

User class	Profile	Primary need
End User	Non-technical; web browser only	Verdict label, confidence score, and plain-Bahasa explanation
Security Researcher	Audio forensics background; needs raw scores	Grad-CAM overlay + per-band attribution percentages
System Developer	Full technical access	Model retraining, API integration, deployment management
Academic Reviewer	ML/AI domain knowledge	Architecture documentation, benchmark results, explainability quality

2.4 Operating Environment

- **Training:** Google Colab Free Tier (Tesla T4 / K80, 12 GB VRAM, 12-hour session limit; gradient checkpointing required)
- **Inference / deployment:** HuggingFace Spaces free tier — 2 vCPU, 16 GB RAM, CPU-only at serving time
- **Client:** Any modern web browser (Chrome 100+, Firefox 100+, Safari 15+)
- **ML stack:** PyTorch 2.x, torchvision, torchaudio, Librosa 0.10.x, pytorch-grad-cam
- **App framework:** Gradio 4.x or Streamlit 1.x; ONNX Runtime mandatory for CPU-optimised inference

> **[Assumption]** *HuggingFace Spaces free tier provides CPU-only inference. All latency targets in Section 4 are calibrated for 2 vCPU / no GPU.*

Section 3
System Features — Functional Requirements

3. System Features — Functional Requirements

Functional requirements are partitioned by modality, following the Architectural Synthesis in Section 0. Each sub-section maps to one technical domain. The 'Category' column in each table identifies the modality tag for traceability.

3.1 Audio / Speech Processing Module [FR-AUD]

Modality: Audio / Speech
DSP pipeline — waveform → spectrogram tensor

3.1.1 Feature description

The Audio module is the sole entry point for raw data. It normalises heterogeneous audio files into a deterministic fixed-size float32 tensor. Correctness here is binary: any deviation from the output contract silently corrupts all downstream stages.

3.1.2 Requirements

ID	Description	Priority	Category
FR-AUD-001	The system SHALL accept audio file uploads in WAV and FLAC formats via the web interface.	**SHALL**	Audio
FR-AUD-002	The system SHALL resample all inputs to 16,000 Hz using high-quality resampling (Kaiser-best window or equivalent).	**SHALL**	Audio
FR-AUD-003	The system SHALL downmix multi-channel audio to mono by averaging all channels before feature extraction.	**SHALL**	Audio
FR-AUD-004	The system SHALL enforce a fixed clip duration of 2.0 seconds (32,000 samples). Clips longer than 2 s SHALL be centre-cropped; clips shorter than 2 s SHALL be zero-padded at the right boundary.	**SHALL**	Audio
FR-AUD-005	The system SHALL reject audio files shorter than 0.5 seconds and return a structured JSON error with code AUD-001.	**SHALL**	Audio
FR-AUD-006	The system SHALL extract a Mel Spectrogram with the parameters: n_mels=128, n_fft=2048, hop_length=512, window=Hann. Parameters SHALL be configurable via config.yaml.	**SHALL**	Audio
FR-AUD-007	The system SHALL convert the power spectrogram to a decibel scale using librosa.power_to_db(ref=np.max) and normalise to [0, 1] per sample using min-max normalisation.	**SHALL**	Audio

ID	Description	Priority	Category
FR-AUD-008	The system SHALL produce a [3, 224, 224] float32 tensor by stacking the normalised single-channel spectrogram across three channels and resizing.	**SHALL**	Audio
FR-AUD-009	The preprocessing module SHALL expose a batch processing API accepting a list of file paths, enabling offline dataset processing.	**SHOULD**	Audio
FR-AUD-010	The system SHOULD log sample rate, duration, and channel count of each processed file to a structured JSON log.	**SHOULD**	Audio
FR-AUD-011	MP3 and OGG formats MAY be accepted as secondary input formats using torchaudio backend.	**MAY**	Audio

3.2 Computer Vision Module [FR-CV]

Modality: Computer Vision
CNN classification + Grad-CAM explainability

3.2.1 Feature description

The CV module is the core predictive engine. It ingests the spectrogram tensor from the Audio module and produces both a prediction (bonafide / spoof with confidence) and an explanation artefact (Grad-CAM saliency heatmap + per-band attribution). The CV module is entirely self-contained — it has no dependency on the NLP module.

3.2.2 Classification requirements

ID	Description	Priority	Category
FR-CV-001	The system SHALL use an EfficientNet-B4 backbone pretrained on ImageNet-1k (torchvision) as the base feature extractor.	**SHALL**	CV — Model
FR-CV-002	The system SHALL replace the EfficientNet-B4 classification head with a two-unit linear layer outputting logits for {bonafide=0, spoof=1}.	**SHALL**	CV — Model
FR-CV-003	Training SHALL proceed in two phases: (1) frozen backbone for 5 epochs (head only); (2) fine-tuning top-N layers at lr ≤ 1e-4. N SHALL be determined by validation EER monitoring.	**SHALL**	CV — Training
FR-CV-004	The system SHALL apply sigmoid activation to the output logit to produce a calibrated probability in (0, 1). Prediction = bonafide if score ≥ 0.5.	**SHALL**	CV — Inference
FR-CV-005	Training SHALL use Binary Cross-Entropy loss with class weights inversely proportional to class frequency.	**SHALL**	CV — Training
FR-CV-006	Training SHALL apply SpecAugment (time mask ≤ 10%, frequency mask ≤ 10%), random time-shift (±0.1 s), and additive Gaussian noise (SNR ≥ 20 dB).	**SHOULD**	CV — Training

ID	Description	Priority	Category
FR-CV-007	The system SHALL serialise the best checkpoint (by validation AUC-ROC) to a .pth file and upload to HuggingFace Model Hub.	**SHALL**	CV — Persistence
FR-CV-008	The system SHALL achieve EER ≤ 10% and AUC-ROC ≥ 0.90 on the FoR held-out test set as minimum acceptance criteria.	**SHALL**	CV — Quality
FR-CV-009	EfficientNet-B0 MAY be used as a lightweight baseline during Week 1 prototyping before B4 fine-tuning begins.	**MAY**	CV — Model

3.2.3 Grad-CAM / XAI requirements

ID	Description	Priority	Category
FR-CV-010	The system SHALL implement Grad-CAM applied to the activations of the final MBConv convolutional block of EfficientNet-B4 (layer: model.features[-1]).	**SHALL**	CV — XAI
FR-CV-011	The system SHALL use the pytorch-grad-cam library (jacobgil/pytorch-grad-cam) for Grad-CAM computation. The target layer name SHALL be locked in config.yaml.	**SHALL**	CV — XAI
FR-CV-012	The system SHALL overlay the normalised Grad-CAM heatmap onto the original Mel Spectrogram using a jet colormap with alpha blending (α = 0.5), returning the result as a PNG image.	**SHALL**	CV — XAI
FR-CV-013	The system SHALL decompose the Grad-CAM saliency map into four acoustic frequency bands using the Mel filter bank's frequency-to-bin mapping (not a naive linear slice): (1) 0–500 Hz, (2) 500 Hz–2 kHz, (3) 2–4 kHz, (4) 4–8 kHz.	**SHALL**	CV — XAI
FR-CV-014	Per-band attribution percentages SHALL be normalised via a Softmax operation (or equivalent normalisation) so that the four band values sum to exactly 100%, eliminating visual ambiguity in the attribution display. The normalised percentages SHALL be passed as structured data to the NLP module prompt builder.	**SHALL**	CV — XAI
FR-CV-015	The XAI module SHALL complete Grad-CAM computation in ≤ 3 seconds on CPU hardware for a single clip.	**SHALL**	CV — XAI
FR-CV-016	The system SHOULD expose the raw saliency matrix as a JSON endpoint for developer access.	**SHOULD**	CV — XAI

3.3 NLP Module [FR-NLP]

Modality: NLP
Qwen 2.5 prompt engineering + English explanation generation

3.3.1 Feature description

The NLP module is the only externally-dependent component in the pipeline. It receives structured outputs from the CV module — label, confidence, band attributions — and produces a natural-language explanation paragraph in English via the Qwen 2.5 API. The UI SHALL display Stage 2 (CV) results independently of Stage 3 completion, isolating users from API latency.

3.3.2 Requirements

ID	Description	Priority	Category
FR-NLP-001	The system SHALL construct a structured prompt containing: (a) predicted label; (b) confidence score; (c) per-band Grad-CAM attribution percentages; (d) instruction to produce a 3–5 sentence explanation in English referencing the frequency band findings.	**SHALL**	NLP — Prompt
FR-NLP-002	The system SHALL invoke the Qwen 2.5 model via its designated API endpoint with a 30-second timeout.	**SHALL**	NLP — Integration
FR-NLP-003	If the Qwen 2.5 API call fails or exceeds the timeout, the system SHALL fall back to a deterministic rule-based template explanation and SHALL surface a warning badge ('AI explanation unavailable') in the UI.	**SHALL**	NLP — Resilience
FR-NLP-004	The generated explanation SHALL be in grammatically correct English and SHALL explicitly reference the highest-attribution frequency band identified by Grad-CAM.	**SHALL**	NLP — Quality
FR-NLP-005	API credentials SHALL be stored as HuggingFace Spaces environment secrets and SHALL NOT appear in any client-side code, logs, or public repository.	**SHALL**	NLP — Security
FR-NLP-006	The NLP module SHALL be invoked asynchronously — the CV verdict SHALL be displayed to the user before NLP completion.	**SHALL**	NLP — Architecture
FR-NLP-007	If Qwen 2.5 API is unavailable, the system SHOULD substitute Gemma-3 as a fallback LLM provider.	**SHOULD**	NLP — Resilience
FR-NLP-008	The system SHOULD cache NL explanations for identical (label, confidence_bucket, top_band) tuples to reduce redundant API calls during demonstration.	**SHOULD**	NLP — Performance
FR-NLP-009	The system MAY support English-language explanations as a secondary output language, selectable via a UI toggle.	**MAY**	NLP — I18N

3.4 Deployment & UI Module [FR-DEP]

Module: Deployment
Gradio / Streamlit web interface on HuggingFace Spaces

3.4.1 Feature description

The deployment module packages the complete four-stage pipeline into a publicly accessible web application. The interface must be usable by all four user classes without local environment setup or authentication.

3.4.2 Requirements

ID	Description	Priority	Category
FR-DEP-001	The system SHALL be deployed as a Gradio 4.x or Streamlit 1.x application on HuggingFace Spaces free tier.	**SHALL**	Deploy

ID	Description	Priority	Category
	The deployment SHALL require no user authentication to access.		
FR-DEP-002	The UI SHALL expose an audio file upload widget accepting .wav and .flac files up to 20 MB.	**SHALL**	Deploy
FR-DEP-003	Upon upload submission, the UI SHALL display the audio waveform visualisation and Mel Spectrogram image while Stage 2 inference is running.	**SHALL**	Deploy
FR-DEP-004	The verdict panel SHALL display: (a) label badge (Bonafide / Spoof); (b) calibrated confidence score (%); (c) horizontal progress bar; — all as soon as Stage 2 inference completes, before NLP finishes.	**SHALL**	Deploy
FR-DEP-005	The UI SHALL display the Grad-CAM saliency overlay image alongside the four-band attribution bar chart.	**SHALL**	Deploy
FR-DEP-006	The UI SHALL display the NLP explanation in a dedicated panel labelled 'AI-generated explanation (English)'. A loading spinner SHALL appear while the API call is pending.	**SHALL**	Deploy
FR-DEP-007	End-to-end wall time from upload submission to full result display SHALL not exceed 15 seconds on HuggingFace Spaces CPU.	**SHALL**	Deploy
FR-DEP-008	The application SHALL include at least 2 bonafide and 2 spoof sample audio files enabling one-click demonstration without user upload.	**SHOULD**	Deploy
FR-DEP-009	The application SHALL include an 'About' section with project description, dataset citation, model architecture summary, and team credits.	**SHOULD**	Deploy
FR-DEP-010	The system SHALL support ONNX export of the EfficientNet-B4 model for reduced CPU inference latency. ONNX Runtime SHALL be used at serving time to guarantee inference performance on HuggingFace Spaces CPU.	**SHALL**	Deploy

Section 4
Non-Functional Requirements

4. Non-Functional Requirements

NFRs are structured by the ISO/IEC 25010 quality model, adapted for this prototype scope. Each row specifies the quality attribute, sub-attribute, requirement statement, and measurable target.

4.1 Performance

Attribute	Sub-attribute	Requirement	Metric / Target
Performance	Preprocessing latency	Audio DSP module SHALL process one 2-second clip on CPU	≤ 500 ms
Performance	CV inference latency	EfficientNet-B4 forward pass on CPU (no ONNX)	≤ 3,000 ms
Performance	Grad-CAM compute time	Saliency map + band attribution on CPU	≤ 3,000 ms
Performance	NLP API latency	Qwen 2.5 API time-to-first-byte	≤ 8,000 ms
Performance	End-to-end wall time	Upload → full result on HuggingFace Spaces CPU	≤ 15,000 ms
Performance	Training throughput	EfficientNet-B4 fine-tuning on Colab T4 GPU	≥ 60 samples/sec
Performance	Model checkpoint size	Serialised .pth file on HF Model Hub	≤ 250 MB

> **[Assumption]** *ONNX Runtime (FR-DEP-010) is mandatory for CV inference on HuggingFace Spaces CPU. The target CV inference latency is ≤ 1,500 ms with ONNX Runtime. This target is binding and SHALL be verified during deployment testing.*

4.2 Accuracy / Model Quality

Attribute	Sub-attribute	Requirement	Metric / Target
Accuracy	EER	Equal Error Rate on FoR held-out test set	≤ 10%
Accuracy	AUC-ROC	Area under ROC curve, FoR test set	≥ 0.90
Accuracy	F1-Score (macro)	Harmonic mean of precision and recall	≥ 0.88
Accuracy	Overall accuracy	Correct classifications / total samples	≥ 90%
Accuracy	ECE	Expected Calibration Error of output probabilities [Assumption]	≤ 0.05
Accuracy	XAI faithfulness	Grad-CAM deletion AUC (proxy metric) [Assumption]	≥ 0.65

4.3 Security

Attribute	Sub-attribute	Requirement	Metric / Target
Security	Credential storage	API keys and HF tokens SHALL be stored as HF Spaces secrets	Never in code/logs
Security	Data retention	Uploaded audio files SHALL be processed in-memory only; no disk persistence after inference	0 bytes persisted
Security	PII	The system SHALL NOT collect, log, or transmit user PII	Zero PII logged
Security	Error disclosure	Internal stack traces SHALL NOT be exposed in end-user error messages	UI shows generic message
Security	Dependency audit	All Python dependencies SHALL be pinned in requirements.txt [Assumption]	Reproducible build

4.4 Reliability

Attribute	Sub-attribute	Requirement	Metric / Target
Reliability	Graceful degradation	Qwen 2.5 API failure SHALL trigger rule-based fallback; CV results remain available	Stage 2 always shown
Reliability	Error handling	Malformed / corrupt audio files SHALL be handled without application crash	Structured error returned
Reliability	Checkpoint resilience	Model checkpoint SHALL be saved to HF Hub every epoch to survive Colab session expiry	Per-epoch persistence
Reliability	Cold-start tolerance [Assumption]	HF Spaces cold-start ≤ 30 s; warm-up ping RECOMMENDED	< 30 s cold start

4.5 Scalability

Attribute	Sub-attribute	Requirement	Metric / Target
Scalability	Stateless inference	The inference pipeline SHALL be stateless to permit future horizontal scaling	No shared mutable state
Scalability	ONNX portability	The model SHALL be exported to ONNX and served via ONNX Runtime for CPU inference on HuggingFace Spaces	ONNX export verified; inference latency ≤ 1,500 ms on CPU
Scalability	Concurrent users [Assumption]	Multi-user concurrency and request queuing are out of scope for v2.0	N/A — single-user prototype

4.6 Maintainability

Attribute	Sub-attribute	Requirement	Metric / Target
Maintainability	Module separation	Audio DSP, CV, NLP, and Deployment SHALL be separate Python modules with defined interfaces	4 discrete modules

Attribute	Sub-attribute	Requirement	Metric / Target
Maintainability	Configuration centralisation	All hyperparameters SHALL be in config.yaml; no magic numbers in source code	Single config file
Maintainability	Source control	All code SHALL be maintained in a public GitHub repository with README, install, train, and deploy instructions	Public GitHub repo
Maintainability	Code comments	Non-obvious decisions (Grad-CAM target layer, Mel bin mapping) SHALL include inline explanations	Inline docstrings

Section 5
Open Questions & Technical Gaps

5. Open Questions

These ambiguities were not resolved in the brainstorming notes. All items marked HIGH must be resolved before the corresponding implementation phase begins. Items MEDIUM must be resolved before Week 3.

#	Owner	Question	Priority
Q1	Both	SEA-LION access mode: RESOLVED/CLOSED — Qwen 2.5 (Alibaba Cloud API) is adopted as the primary LLM for English explanation generation. SEA-LION dependency is removed.	**HIGH**
Q2	Both	FoR dataset variant: RESOLVED/CLOSED — The for-2sec split (2-second clips) is adopted. FR-AUD-004 has been updated to 2.0 seconds (32,000 samples), resolving the prior conflict with the 4-second assumption.	**HIGH**
Q3	Both	EfficientNet variant: has the team verified that B4 training with batch_size ≥ 16 is feasible under Colab Free VRAM? Gradient checkpointing may be required.	**HIGH**
Q4	Ferel	Grad-CAM target layer: what is the exact PyTorch attribute path for the final MBConv block in torchvision EfficientNet-B4? Misidentification silently produces wrong saliency.	**HIGH**
Q5	Safa	Frequency band mapping: band attributions must use the Mel filter bank's frequency-to-bin mapping (FR-CV-013), not a naive row-slice. Has this mapping logic been validated on a test spectrogram?	**MEDIUM**
Q6	Both	Deployment framework: Gradio or Streamlit? Must be locked before Week 4 UI work begins. Gradio recommended for native audio widget support.	**MEDIUM**
Q7	Both	Evaluation protocol: official ASVspoof EER scoring script or scikit-learn implementation? The former is required if submitting to academic conferences.	**LOW**

Appendix A — Glossary

Bonafide	Authentic human speech, not synthesised or voice-converted.
Spoof / Deepfake	Synthetically generated or voice-converted speech intended to impersonate.
EER	Equal Error Rate — where false acceptance rate = false rejection rate.
Mel Spectrogram	2D time-frequency audio representation on the perceptual Mel scale.
STFT	Short-Time Fourier Transform — decomposes audio into time-frequency bins.
MBConv	Mobile Inverted Bottleneck Convolution — core block in EfficientNet.
SE block	Squeeze-and-Excitation — channel-attention mechanism in each MBConv.
Grad-CAM	Gradient-weighted Class Activation Mapping — CNN saliency via gradient backprop.
Qwen 2.5	Qwen 2.5 large language model by Alibaba Cloud — used for English explanation generation.
FoR Dataset	Fake-or-Real audio dataset by M. Abdel-Dayem (Kaggle).
SpecAugment	Data augmentation applying time and frequency masking to spectrograms.
Transfer learning	Using ImageNet-pretrained weights as starting point for fine-tuning.
HF Spaces	HuggingFace Spaces — free-tier cloud ML app deployment platform.
IEEE 830	IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications.
XAI	Explainable Artificial Intelligence.

Appendix B — Requirement Traceability Matrix

Maps each functional requirement to its source, modality, and the open question (if any) that must be resolved before implementation.

FR ID	Modality	Priority	Implementation dependency	Blocks on
FR-AUD-001–008	Audio	SHALL	librosa 0.10.x + torchaudio installed	Q2 (dataset variant — Resolved)
FR-CV-001–009	Computer Vision	SHALL	torchvision pretrained weights; Colab T4 GPU available	Q3 (VRAM), Q4 (layer)
FR-CV-010–016	Computer Vision	SHALL	pytorch-grad-cam; Mel bin-to-Hz mapping validated	Q4, Q5
FR-NLP-001–009	NLP	SHALL/SHOULD	Qwen 2.5 API key or fallback LLM API key available	Q1 (API access — Resolved)
FR-DEP-001–010	Deployment	SHALL/SHOULD	Gradio or Streamlit framework locked	Q6 (framework)

End of Document — DSDBA SRS v2.1 — IEEE 830 — KCVanguard Final Project, ITS Informatics — 2026-03-17